Multiattention Joint Convolution Feature Representation With Lightweight Transformer for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is currently a hot topic in the field of remote sensing. The goal is to utilize the spectral and spatial information from HSI to accurately identify land covers. Convolution neural network (CNN) is a powerful approach for HSI classification. However, CNN has limited ability to capture nonlocal information to represent complex features. Recently, vision transformers (ViTs) have gained attention due to their ability to process non-local information. Yet, under the HSI classification scenario with ultrasmall sample rates, the spectral–spatial information given to ViTs for global modeling is insufficient, resulting in limited classification capability. Therefore, in this article, multiattention joint convolution feature representation with lightweight transformer (MAR-LWFormer) is proposed, which effectively combines the spectral and spatial features of HSI to achieve efficient classification performance at ultrasmall sample rates. Specifically, we use a three-branch network architecture to extract multiscale convolved 3D-CNN, extended morphological attribute profile (EMAP), and local binary pattern (LBP) features of HSI, respectively, by taking full exploitation of ultrasmall training samples. Second, we design a series of multiattention modules to enhance spectral–spatial representation for the three types of features and to improve the coupling and fusion of multiple features. Third, we propose an explicit feature attention tokenizer (EFA-tokenizer) to transform the feature information, which maximizes the effective spectral–spatial information retained in the flat tokens. Finally, the generated tokens are input to the designed lightweight transformer for encoding and classification. Experimental results on three datasets validate that MAR-LWFormer has an excellent performance in HSI classification at ultrasmall sample rates when compared to several state-of-the-art classifiers.**

***Index Terms*— Cross attention mechanisms, hyperspectral image (HSI) classification, multiattention, multifeature, transformer.**

I. Introduction

HYPERSPECTRAL images are generated by a spectral imager that scans the target area from dozens to hundreds of consecutive bands [1]. They combine rich spatial information with a stack of spectral information in hundreds of bands [2]. As a result, hyperspectral images (HSIs) contain a wealth of surface information and effectively reflect the properties of the Earth's surface, leading to wide applications in remote sensing [3], national defense [4], medical diagnosis [5], precision agriculture [6], mineral development [7], and so on. For decades, researchers have carried out a series of technical processing work on HSIs [8], [9], [10], [11], [12], [13], [14]. As a fundamental technique, HSI classification predicts the ground category of each pixel in an image based on spectral and spatial information [15]. However, the shortcomings of HSIs, such as high dimensionality, unbalanced sample distribution, high information redundancy, and low spatial resolution, also pose challenges for feature representation in classification tasks [16].

During the last decades, many researchers have devoted themselves to HSI classification using handcrafted feature representation. For example, Morphological Attribute features [17], local binary pattern (LBP) features [18], and Gabor features [19] are typical of these representations utilizing texture information and shape information. For instance, Dalla Mura et al. [20] introduced the extended morphological attribute profiles (EMAPs) mechanism to extract detailed morphological attributes during principal component analysis (PCA) and input the features into a support vector machine (SVM), which enhanced the spectral–spatial representation. Huang et al. [21] used a multikernel representation to extract LBP features and achieved effective utilization of spatial features. Jia et al. [22] used Gabor filters to extract 3D-Gabor features from HSI to fuse spectral information into spatial texture features, and achieved promising results. However, handcrafted features require a strong theoretical background, which must deal with many complex parameters, and it is difficult to obtain deep feature representations [23]. Therefore, the representation and learning capability of handcrafted features are limited.

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Recently, deep neural networks have emerged as the preferred tool for various image classification tasks [24]. Deep neural networks can learn deeper features and obtain more representative image information by designing a reasonable deep structure. HSI classification with deep neural networks can capture deeper spectral–spatial features and significantly improve classification performance. Hu et al. [25] proposed a convolution neural network (CNN) structure for HSI classification, which replaced handcrafted features with deep features and achieved exciting results. Since then, deep neural networks have made a splash in the field of HSI classification. Liu et al. [26] proposed a novel deep convolution feature extraction method using a Siamese CNN that mapped the data to a supervised marginal ranking loss function to extract more discriminative features for the classification task. Roy et al. [27] extracted more precise spectral–spatial features from HSI by overlaying and mixing 3D-CNN with 2D-CNN, which greatly enhanced the combination of spectral and spatial information. In addition, some recent deep learning approaches are also refreshing [28]. Hong et al. [29] proposed a mini-batch graph convolution network that can be directly fused with convolution features, increasing the capability to model HSI global relations. Wang et al. [30] designed a two-channel spectral–spatial fusion capsule generation adversarial network for HSI classification to break the high-dimensional feature size limitation by employing feature information from different channels, thus further improving the accuracy and robustness of classification. Although deep learning methods have achieved absolute success in the field of HSI classification, all these deep networks require a large number of training samples to boost the network, otherwise serious under-fitting will occur. Therefore, for HSI classification tasks at ultrasmall sample rates, it is not easy for ordinary CNN to demonstrate their strengths in exploiting effective deep features.

To address the HSI classification problem in the case of insufficient samples, researchers have proposed various solutions in recent years, such as multiscale feature extraction, multifeature classification fusion, and multimodal classification fusion. Sun et al. [31] proposed an multistructure KELM with attention fusion strategy (MSAF-KELM) which extracted different features at multiple scales of HSI into multiple kernel extreme learning machines to perform classification simultaneously, and finally implemented the fusion of multiple branches using a self-attention mechanism. MSAF-KELM achieved efficient classification at ultrasmall sample rates, overcoming the extreme case of insufficient training samples. Hong et al. [32] creatively broke through the limitation of remote sensing image types and realized HSI classification with the assistance of multiple types of remote sensing images. Zhu et al. [33] extracted features on HSI from three stereo feature perspectives, namely 3-D morphology, 3-D LBP, and the proposed 3D-surface feature, respectively, and were able to maintain excellent classification results even with a very small number of samples. Multiple feature sources can effectively alleviate the problem of insufficient samples due to the high dimensionality of HSI. Thus, there are three types of HSI features used in our proposed multiattention joint convolution feature representation with lightweight transformer (MAR-LWFormer): original HSI 3D-CNN features, EMAP features, and LBP features. Among them, the original image 3D-CNN features contain rich spectral information and are combined with the spatial features, e.g., EMAP and LBP features, to fully exploit the feature information under ultrasmall sample rates from multiple perspectives.

For the multifeature approach in the HSI classification task, although it solves the under-fitting problem caused by insufficient training samples, too much feature information and network branches tend to bring challenges to the fusion work. In computer vision tasks, we often use attention mechanisms to eliminate some irrelevant information and maximize the retention of the focus on which the task should be focused. Similar ideas can be used in multifeature methods. The input of multiple features inevitably brings a lot of redundant information while obtaining more effective information. The attention mechanism can simulate the attention mechanism of human eyes to select useful information as much as possible within limited resources. Sun et al. [31] proposed a weighted self-attention fusion strategy (WSAFS) in MSAF-KELM to retain the more important classification results among multiple classifiers, which effectively improved the fusion of multiple features. Paoletti et al. [34] proposed an attention-guided capsule network that used a detailed feature extractor coupled with an attention mechanism to identify the most representative visual parts of the HSI. Both the computational efficiency and the classification accuracy were improved. Zhu et al. [35] designed a spectral–spatial attention module to emphasize effective spectral and effective spatial pixels, which obtained excellent classification accuracy. In summary, the attention mechanism has been shown to be extremely powerful for improving classification performance. In MAR-LWFormer, we propose a cross-feature attention fusion (CFAF) module and a cross-dimensional spectral enhancement attention (CSE-A) module for the generation of joint convolution features. By cross-fusing two types of spatial features and performing enhanced spectral attention calculations on the features, the original convolution is able to significantly improve the spectral–spatial representation. This approach effectively combines information in the spectral–spatial domain and focuses on important regions, resulting in complementary and informative features. As a result, the joint convolution representation features are generated, maximizing the utilization of limited information and greatly enhancing the feature representation capability for ultrasmall samples.

Recently a model called vision transformer (ViT) [36] has been sweeping the field of computer vision. The original form of ViT transformer consists of several self-attention mechanisms, which were originally applied in the field of NLP. The HSI classification model using ViT generally converts image features into the form of tokens and learns each token interactively several times within the structure to fully exploit the spectral–spatial logic in HSI. Hong et al. [37] proposed a SpectralFormer which learned the feature representation of neighboring spectra and reduced the loss error when the features were propagated across layers. Xue et al. [38] designed a local transformer with a spatial partition restoration network,

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which obtained spatial attention weights by measuring the similarity between related pixels and used the spatial partition restoration module to achieve global modeling of hyperspectral data. Sun et al. [39] proposed a spectral–spatial feature tokenization transformer (SSFTT), which mixed and stacked a set of 3D-CNN with 2D-CNN to generate a Gaussian distribution token, and then those tokens were fed to the transformer to fully exploit the deep semantic features in HSI. However, the transformer requires a certain amount of data to model the global information, and using the transformer encoder at ultrasmall sample rates may result in a poor attention effect. In addition, a token as the input form of the transformer has the property of being lightweight, and too large tokens will lead to extremely high computation and huge parameters. Therefore, in our proposed MAR-LWFormer, we have introduced an explicit feature attention tokenizer (EFA-tokenizer) that performs feature attention computation during the compression of multifeature information. This is done to maximize the effective spectral–spatial information retained in the flat token while eliminating redundant information between features and reducing computational complexity. Moreover, we have improved the multihead self-attention mechanism of the original ViT and designed a lightweight pooling mixer to effectively reduce the computational complexity associated with multifeature redundancy.

In summary, the MAR-LWFormer method proposed in this article has the following main contributions.

1) The MAR-LWFormer proposes a multibranch network to exploit multiple and multiscale spectral–spatial information of HSI. Specifically, it enhances the spectral–spatial feature representation of HSI by a multiscale 3D-CNN and extracts the convolved EMAP and LBP features by 2D-CNNs. In this way, it fully exploits the representation information of a single pixel from multiple perspectives and effectively improves the classification accuracy at ultrasmall sample rates.
2) The MAR-LWFormer proposes a CFAF module to achieve effective fusion of three types of convolution data and design a CSE-A module to improve the coupling of spectral–spatial features.
3) The MAR-LWFormer proposes an EFA-tokenizer and a lightweight transformer encoder, which compresses multifeature information into tokens while eliminating redundant information between features and keeping the computational effort minimal on transformer operations.
4) The MAR-LWFormer can achieve better classification performance than several classical and SOTA methods at ultrasmall sample rates. Experiments on all three famous datasets show excellent performance of the proposed MAR-LWFormer classifier.

The remainder of this article is organized as follows. Section II reviews the knowledge related to the algorithm in depth and details the proposed MAR-LWFormer. Section III gives a lot of experiments and analyses. Section IV summarizes this article and puts forward some opinions and suggestions for future research.

II. Materials and Methods

Fig. 1 illustrates the general framework diagram for the HSI classification using the proposed MAR-LWFormer. It contains three main components: the spectral-spatial multifeature convolution extraction stage, the multiattention joint feature representation stage, and the transformer encoder with the EFA-tokenizer.

A. Spectral-Spatial Multifeature Convolution Extraction

1) Multiscale Spectral–Spatial Convolution Processing: HSI data contain abundant spectral–spatial information, based on which ground objects can be effectively distinguished. Therefore, maximizing the distinction between spectral and spatial information in course of feature extraction is particularly important for the classification task. Specifically, we employ a multiscale 3D-CNN to extract both spectral and spatial information from multiple perspectives. 3D-CNN uses 3-D convolution kernels to convolute the HSI cube data along the spectral domain, but it brings the problem of detail loss with the increasing depth of the network. Therefore, we set up three types of convolution kernels separately with different sizes to extract multiscale spectral–spatial information. To avoid the introduction of excessive parameters and prevent the overfitting phenomenon, we perform the PCA to reduce the dimensionality of the original HSI before the 3-D convolution operation, eliminating the invalid redundant bands and retaining the most dominant part of the bands. Let the input HSI be $\mathbf{I} \in \mathbf{R}^{m\times n\times l}$, where $m \times n$ is the spatial size and l is the number of spectral bands, and the l-band spectra is downscaled to the b-band ones by PCA, i.e., $\mathbf{I}_{\text{pca}} \in \mathbf{R}^{m\times n\times b}$.

Before performing the multiscale 3D-CNN operation on the HSI, it is first divided into multiple neighboring 3-D-patches $\mathbf{P} \in \mathbf{R}^{s\times s\times b}$. Subsequently, the multiscale 3D-CNN operation is performed, and the multiscale propagation operation for the jth feature cube at the (x, y, z) position on the ith layer is given by

$$\begin{aligned}
\alpha_{ij}^{xyz} &= \Phi\left(\sum_m \sum_{h=0}^{HL_{i-1}} \sum_{w=0}^{WL_{i-1}} \sum_{r=0}^{RL_{i-1}} w_{ijm}^{hwr}\, \alpha_{(i-1)m}^{(x+h)(y+w)(z+r)} + b_{ij}\right)\\
\beta_{ij}^{xyz} &= \Phi\left(\sum_m \sum_{h=0}^{HM_{i-1}} \sum_{w=0}^{WM_{i-1}} \sum_{r=0}^{RM_{i-1}} w_{ijm}^{hwr}\, \beta_{(i-1)m}^{(x+h)(y+w)(z+r)} + b_{ij}\right)\\
\delta_{ij}^{xyz} &= \Phi\left(\sum_m \sum_{h=0}^{HS_{i-1}} \sum_{w=0}^{WS_{i-1}} \sum_{r=0}^{RS_{i-1}} w_{ijm}^{hwr}\, \delta_{(i-1)m}^{(x+h)(y+w)(z+r)} + b_{ij}\right)\\
v_{ij}^{xyz} &= \text{cat}\left(\alpha_{ij}^{xyz}, \beta_{ij}^{xyz}, \delta_{ij}^{xyz}\right) \otimes \text{Filter}_{1\times1\times1} + \beta_{ij}^{xyz} \qquad (1)
\end{aligned}$$

where m refers to the feature cube associated with the jth feature cube, HL × WL × RL, HM × WM × RM, and HS × WS × RS are the three types of convolution kernel sizes, w and b are the weight parameter and bias parameter, Φ is the activation function. α_{ij}^{xyz}, β_{ij}^{xyz}, and δ_{ij}^{xyz} are the convolution features extracted under different angles, and they are cascaded for convolution fusion to further enrich the content of the features and extract deeper semantic information. We have provided detailed experiments in Appendix A of the Supplementary Material to investigate the impact of the

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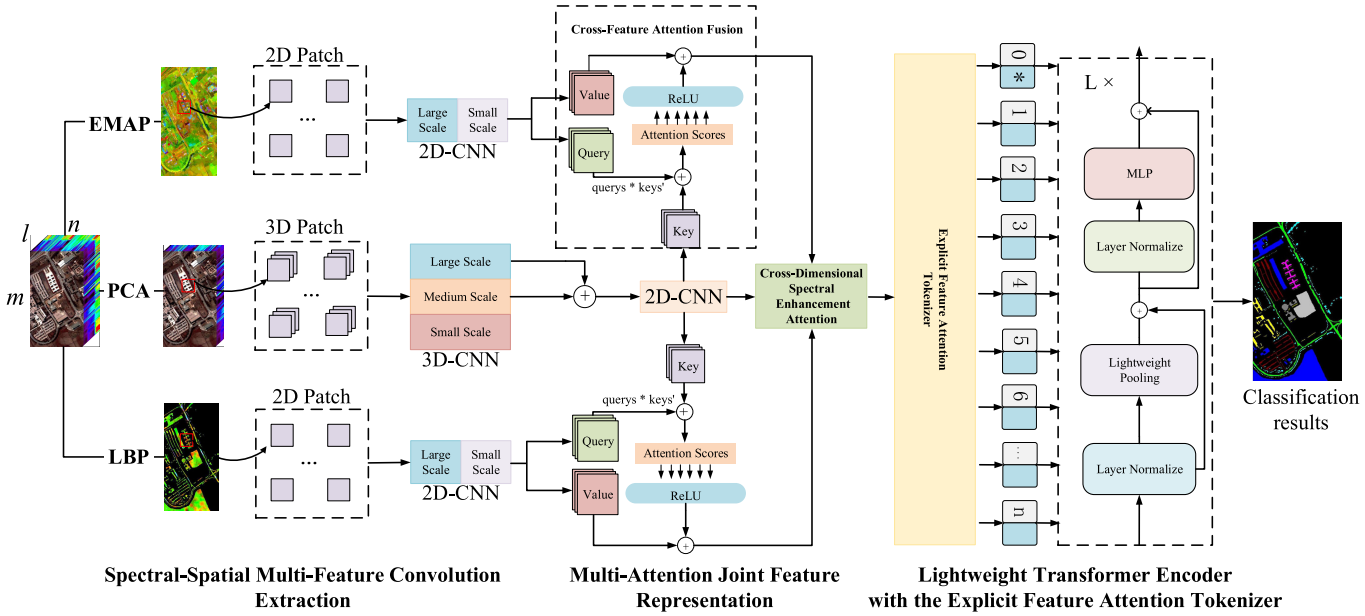


Fig. 1. Main architecture of our proposed MAR-LWFormer algorithm.

multiscale convolution fusion method on the accuracy results in comparison to other fusion methods. To realize the fusion of 3D-CNN with EMAP, LBP features better and enhance the representation of spatial features, we then extract 2D-CNN features from the above feature results to reduce the redundancy of spectral and spatial information and ensure the stability of the next multifeature fusion work. The multiscale propagation operation for the jth feature map at the (x, y) position on the ith layer is given by

$$v_{ij}^{xy} = \Phi\left(\sum_{m}\sum_{h=0}^{H_{i-1}}\sum_{w=0}^{W_{i-1}} w_{ijm}^{hw} v_{(i-1)m}^{(x+h)(y+w)} + b_{ij}\right) \tag{2}$$

where m refers to the feature map associated with the jth feature map, $H \times W$ is the size of the 2-D convolution kernel, w and b are the weight and bias parameters, respectively, and Φ is the activation function.

2) EMAP and LBP Convolution Processing: The 3D-CNN features of the original image contain both rich spectral and spatial information. To further enhance the representation of spatial information and reduce the probability of the "same category with different spectra" and the "same spectra with different categories" problems in HSI classification, we choose two types of spatial features, EMAP and LBP, to incorporate with the original image convolution features. EMAP features can capture the edge and texture information of the image, while LBP features are effective in representing the local pattern information of the image. These two types of features can significantly improve the deficiencies of the original HSI in terms of their insufficient ability to represent spatial features.

The EMAP feature is a feature map composed of superimposing multiple morphological attributes after obtaining the extended attribute profile (AP) separately. We perform EMAP feature extraction on the reduced-dimensional $\mathbf{I}_{\text{pca}} \in \mathbf{R}^{m\times n\times b}$, and the operation is represented as follows:

$$\text{EMAP} = \left\{\text{EAP}_{a_1}, \ldots, \text{EAP}_{a_i}, \ldots, \text{EAP}_{a_n}\right\} \tag{3}$$

where a_i refers to the ith morphological attribute, such as *area*, *inertia*, *standard deviation*, etc. EAP refers to an extended AP, which is a feature vector generated by stacking morphological APs of individual PCs simultaneously in the PCA process. The arithmetic representation is as follows:

$$\text{EAP} = \{\text{AP}(\text{PC}_1), \text{AP}(\text{PC}_2), \ldots, \text{AP}(\text{PC}_b)\} \tag{4}$$

where the AP operation is to perform multiple attribute thickening and thinning operations on a one-wave image to obtain an AP map. The operation is represented as follows:

$$\text{AP}(f) = \{\phi_1(f), \ldots \phi_n(f), f, \gamma_1(f), \ldots, \gamma_n(f)\} \tag{5}$$

where f refers to the one-wave image, and ϕ and γ denote the attribute thickening and thinning operations, respectively.

Next, we extract the LBP features of HSI. The LBP algorithm achieves the extraction of spatial information of the whole image by describing the differences between each central pixel and local domain pixels in the image. Here we use the circular window LBP algorithm where the radius is 1 to achieve the extraction of texture images of different sizes and frequencies. We operate on the first principal component of the HSI, which is represented as follows:

$$\begin{aligned}\text{LBP}_{(c,P)} &= \sum_{i=0}^{p-1} s_{\text{lbp}}(g_i - g_c)2^i \\ S_{\text{lbp}}(x) &= \begin{cases} 1, & x \geqslant 0 \\ 0, & x < 0 \end{cases}\end{aligned} \tag{6}$$

where c refers to the central pixel, P refers to the number of pixels in the field around c, and g refers to the gray value of the pixel.

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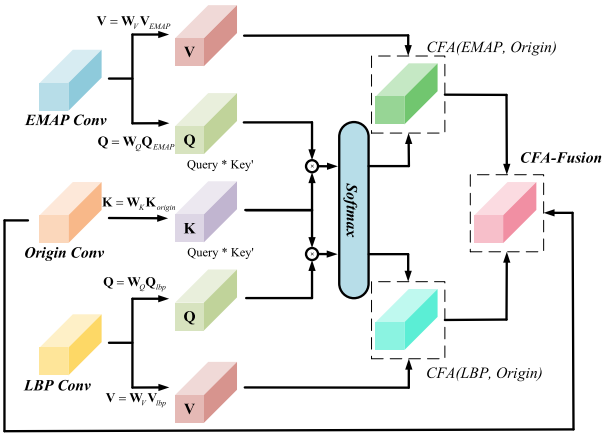


Fig. 2. Illustration of CFAF.

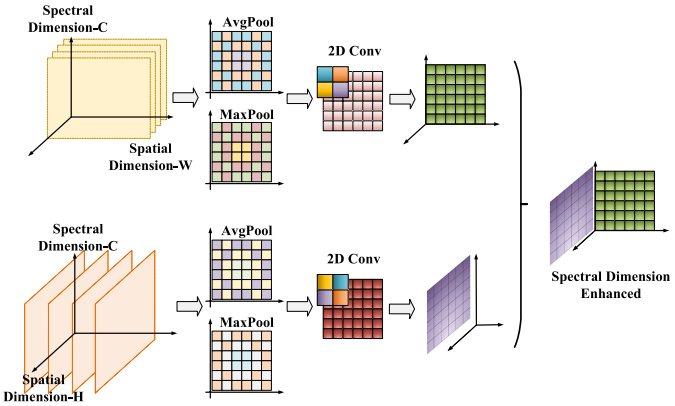


Fig. 3. Illustration of CSE-A.

After extracting the two types of spatial features, we enhance the spatial representation using a multiscale 2D-CNN to convert EMAP and LBP to convolution feature representation

$$\alpha_{ij}^{xy} = \Phi\left(\sum_{m}\sum_{h=0}^{HL_{i-1}}\sum_{w=0}^{WL_{i-1}} w_{ijm}^{hw}\alpha_{(i-1)m}^{(x+h)(y+w)} + b_{ij}\right) \tag{7}$$

$$\beta_{ij}^{xy} = \Phi\left(\sum_{m}\sum_{h=0}^{HS_{i-1}}\sum_{w=0}^{WS_{i-1}} w_{ijm}^{hw}\beta_{(i-1)m}^{(x+h)(y+w)} + b_{ij}\right) \tag{8}$$

$$v_{ij}^{xy} = \text{cat}\left(\alpha_{ij}^{xy}, \beta_{ij}^{xy}\right) \otimes \text{Filter}_{1\times 1} \tag{9}$$

where m refers to the feature map associated with the jth feature map, HL × WL, and HS × WS are different types of convolution kernel sizes, w and b are the weight parameter and bias parameter, respectively, and Φ is the activation function, α_{ij}^{xy} and β_{ij}^{xy} are the convolution features extracted at different angles, and they are cascaded and convoluted for fusion. In addition, we provide detailed information on experiments that explore the impact of the multiscale convolution fusion method of spatial convolution in Appendix A of the Supplementary Material.

B. Multiattention Joint Feature Representation

1) Cross-Feature Attention Fusion: After performing the spectral–spatial multifeature convolution extraction, the EMAP and LBP spatial information is fused into the original HSI features using the CFAF module. Unlike other fusion methods, this module facilitates cross-attentive interaction between the original features and the convolved EMAP and LBP features, ensuring effective combination of spectral-spatial information while eliminating invalid and redundant features.

The process of CFAF is illustrated in Fig. 2. The CFAF module operates on two types of features directly as attention inputs to maximize the retention of the original feature information. We map the EMAP convolution features into query matrix $\mathbf{Q}$ and value matrix $\mathbf{V}$, and the original HSI features into key matrix $\mathbf{K}$ linearly. We provide a detailed explanation for the specific reasons for their combination, in Appendix B of the Supplementary Material. To initialize the learning matrices $\mathbf{W}_Q$, $\mathbf{W}_V$, and $\mathbf{W}_K$ for the three types of attention variables, we use Xavier and ensure they conform to a uniform random value distribution. The cross-feature attention score is obtained by multiplying $\mathbf{Q}$ containing the EMAP feature information with $\mathbf{K}$ containing the original feature information. After the activation function layer assigns the score weights, it is then multiplied with $\mathbf{V}$ containing EMAP feature information for crossfeature fusion interaction. The fusion interaction of the LBP convolution features and the original features is also as described above. The specific operational flow is as follows:

$$\begin{aligned}\mathbf{Q} &= \mathbf{W}_Q\mathbf{Q}_{\text{emap/lbp}}\\ \mathbf{V} &= \mathbf{W}_V\mathbf{V}_{\text{emap/lbp}}\\ \mathbf{K} &= \mathbf{W}_K\mathbf{K}_{\text{origin}}\end{aligned} \tag{10}$$

$$\begin{aligned}&\text{CFA(EMAP/LBP, Origin)}\\ &\quad= \text{softmax}\left(\frac{\mathbf{Q}_{\text{emap/lbp}}\mathbf{K}^T{}_{\text{origin}}}{\sqrt{d_k}}\right)\mathbf{V}_{\text{emap/lbp}}\end{aligned} \tag{11}$$

where d_k is the number of columns of $\mathbf{Q}$, $\mathbf{K}$, and CFA is the cross-feature attention network. We interact the original features with both convolved EMAP and LBP features, and then add the two types of results directly with the original features to obtain the final joint convolution features. The specific operation flow is as follows:

$$\text{CFAF} = \text{CFA(EMAP, Origin)} + \text{CFA(LBP, Origin)} + \text{Origin}. \tag{12}$$

2) Cross-Dimensional Spectral Enhancement Attention: After the EMAP and LBP features are fused into the original HSI features, the spatial information is greatly abundant and the spatial feature representation capability is enhanced. However, whether the original spectral features match with the enhanced spatial features still affects the final classification results. Therefore, we have designed a CSE-A module to interactively enhance the spectral information from both the vertical and horizontal directions of the spatial domain.

The specific process of CSE-A is illustrated in Fig. 3. CSE-A establishes dependency relationships between the spectral dimension C and spatial dimensions H and W through rotation operations, which calculate attention weights for the spectral and spatial domains, respectively. Then the two types of relational weights are aggregated to realize the interaction

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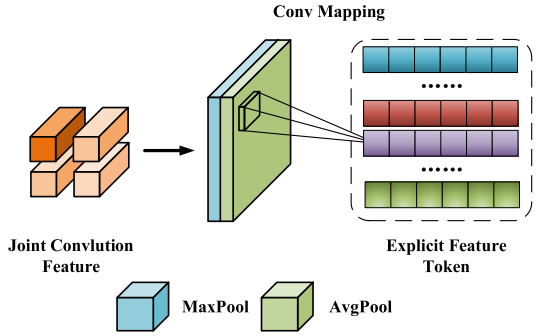


Fig. 4. Illustration of EFA-tokenizer.

and enhance the spectral feature representation. CSE-A is composed of two branches, and the first branch is the spectral dimension C and the spatial dimension H interaction branch. The joint convolution features are converted to $W \times H \times C$ feature maps by rotating 90° counterclockwise along the H-axis, i.e., the information in the H and C dimensions is mapped to the W dimension. Then, average pooling and maximum pooling are performed in the W dimension. Specifically, average pooling compresses the spatial–spectral information of the region into a scalar, which helps smooth the image data and reduce the impact of image noise, thereby facilitating further feature fusion. On the other hand, max pooling preserves the maximum pixel in the spatial and spectral dimensions of the region, discarding a large amount of redundant feature information. Finally, the attention weights between H dimension and C dimension are obtained after the activation function and multiplied by the original joint convolution features to obtain the $C \times H \times W$ dimension features. The other branch is the spectral dimension C and spatial dimension W interaction branch, which is the joint convolution feature rotated 90° counterclockwise along the W-axis to $H \times C \times W$ feature, and the attention weight is calculated in the same way as the first branch. After the two branches are calculated, they are averaged and aggregated to realize the interaction between spectral and spatial dimensions.

C. Lightweight Transformer Encoder With the EFA-Tokenizer

The raw transformer is mainly used for NLP tasks, which takes each word in a sentence as a token and inputs it to the self-attention mechanism for computation. In the ViT, the image is segmented into multiple patches, and each patch is input as a token for the self-attention calculation. The usual tokens are obtained directly through the H and W dimensions of a patch, which can lead to the inclusion of some invalid spatially redundant information for the multifeature fusion task, thus reducing the efficiency of global modeling of the transformer encoder. To alleviate the above problem, we design an EFA-tokenizer and a lightweight pooling token mixer to improve the performance of multifeature classification.

The EFA-tokenizer selects the most important spectral-spatial information to be retained while splitting the joint convolution features into patches and flattening them to maximize the degree of refinement of individual tokens. Specifically, we perform average pooling to retain overall features and maximum pooling to retain locally important features for each patch, respectively. We then map them into 1-D flat features and add them directly to maintain the valid information in the token. The EFA-token generation process is described in Fig. 4. Then, we superimpose the individual EFA-token with the sign cls-token indicating the classification task, add the position token indicating the position information, and input it directly into the transformer encoder. EFA-token minimizes feature redundancy in the transformer encoder for global information interaction.

On the other hand, the computational burden of the transformer encoder is mainly brought by the multihead self-attention mechanism. It is widely believed that the transformer structure achieves excellent results because its token mixer based on the attention mechanism contributes the most. However, the role of the self-attention mechanism is more to serve as a mixing of tokens for interaction [40]. We try to replace the multihead self-attention mechanism with a lightweight pooling mechanism, which is described in Fig. 5. First, the lightweight pooling mechanism implements the token mixer in the most efficient way, which is much more efficient than the multihead attention mechanism in terms of computation. Then, the lightweight pooling mechanism achieves the clustering of each token with neighboring tokens in a straight way, preserving the most abundant global feature information at an ultrasmall sample rate. The data are pooled and mixed along the batch dimension in the following way:

$$
\begin{aligned}
T^{\text{small}}_{:,i,j} &= \frac{1}{P_s \times P_s} \sum_{a,b=1}^{P_s} T_{:,i+a-\frac{P_s+1}{2},i+b-\frac{P_s+1}{2}} - T_{:,i,j} \\
T^{\text{mid}}_{:,i,j} &= \frac{1}{P_m \times P_m} \sum_{a,b=1}^{P_m} T_{:,i+a-\frac{P_m+1}{2},i+b-\frac{P_m+1}{2}} - T_{:,i,j} \\
T^{\text{large}}_{:,i,j} &= \frac{1}{P_l \times P_l} \sum_{a,b=1}^{P_l} T_{:,i+a-\frac{P_l+1}{2},i+b-\frac{P_l+1}{2}} - T_{:,i,j} \\
T^{\text{final}} &= T^{\text{small}}_{:,i,j} \oplus T^{\text{mid}}_{:,i,j} \oplus T^{\text{large}}_{:,i,j}
\end{aligned} \tag{13}
$$

where $T_{:,i,j}$ represents the input data of the multiscale pooling layer, which is the token data that requires interactive mixing. P_s, P_m, and P_l are three-scale pooling layer sizes, which we usually set to a value between 5 and 10 to ensure a balance between performance and complexity.

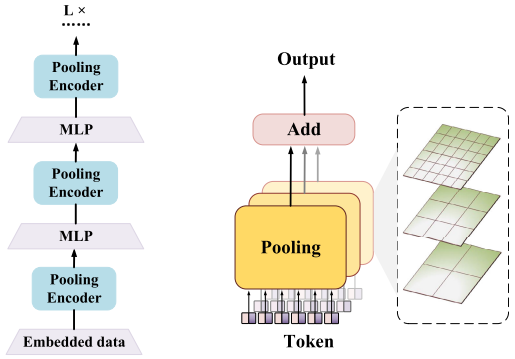
D. Algorithm Summarization for MAR-LWFormer

The overall process of the proposed MAR-LWFormer network is shown in Algorithm 1.

III. Experiment and Analysis

In this section, we conducted experiments on two classical datasets including Pavia University, Salinas, and a novel dataset Xuzhou to demonstrate the effectiveness of our proposed MAR-LWFormer for HSI classification. In addition, several classical and state-of-the-art methods were selected for comparison with MAR-LWFormer, including traditional machine learning methods, i.e., random forest (RF) [41],

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(a) (b)

Fig. 5. Graphical illustration of the lightweight transformer. (a) Overall framework with L encoder blocks. (b) Lightweight pooling encoder mechanism.

Algorithm 1 MAR-LWFormer Network

Input:

HSI data $\mathbf{I} \in \mathbf{R}^{m*n*l}$, ground-truth $\mathbf{Y} \in \mathbf{R}^{m*n}$; PCA bands number b; patch size s; training sample rate $\mu\%$; tokens number t; pooling mixer size p; epochs Δ; batch size = 64; learning rate = 1e-3;.

Output:

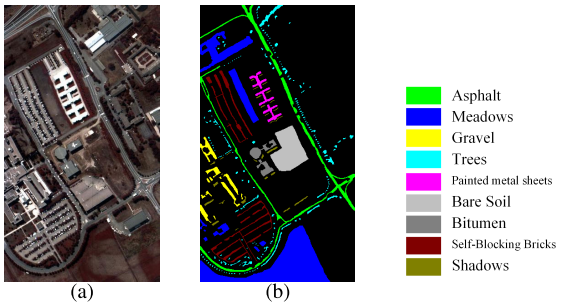
Predicted labels of the test dataset.

Obtain $\mathbf{I}_{pca}$ feature from HSI data and convert them into multiscale convolution features V_{pca}^{xyz} by Eq. (1), (2).
Obtain $\mathbf{I}_{emap}$ by Eq. (5), and $\mathbf{I}_{lbp}$ by Eq. (6) feature from HSI data and convert them into convolution features V_{emap}^{xy}, V_{lbp}^{xy} by Eq. (9).
for $i = 1$ to Δ **do**
    Obtain the spatially enhanced features V_{cfaf}^{xy} by CFAF of Eq. (12) for V_{pca}^{xyz}, V_{emap}^{xy}, and V_{lbp}^{xy}, respectively.
    Perform CSE-A operation on V_{cfaf}^{xy} to achieve enhancement of spectral features.
    Perform EFA-Tokenization transform with feature vectors to generate feature tokens.
    Perform the Lightweight Pooling in Transformer Encoder by Eq. (13).
end for
Input the tokens to the last linear layer.
Use the softmax function to identify the labels.
Use the test dataset with the trained model to get predicted labels.

SVM [42], classic deep learning networks, i.e., 2D-CNN [43], 3D-CNN [44], HybridSN [27], and MSAF-KELM [31], and transformer-based methods, i.e., local transformer with spatial partition restore (SPRLT-Net) [38], and SSFTT [39]. Each method was run on all three datasets, and we set the training sample rate to 0.5% to simulate extreme case classification scenarios to verify the efficient classification performance of the proposed MAR-LWFormer at ultrasmall sample rates.

A. Data Description

1) Pavia University: The Pavia University dataset was imaged by the German reflective optics spectrographic imaging system (ROSIS-03) sensor in 2003 as a HSI dataset for the city of Pavia. It has an image size of 610 × 340 pixels, a spatial resolution of 1.3 m, and contains 103 bands that can be used for classification. The specific label categories and sample set distribution have been shown in Fig. 6 and Table I.

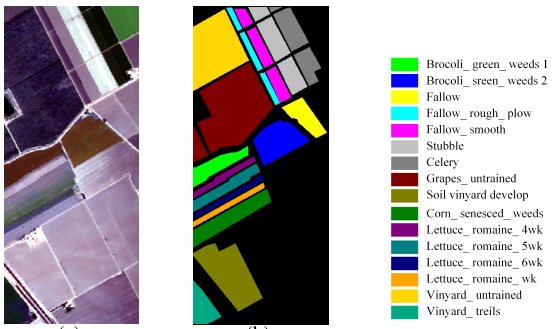


(a) (b)

Fig. 6. Pavia University dataset. (a) False-color map. (b) Ground-truth map.

TABLE I
TRAINING AND TEST SAMPLES IN THE PAVIA UNIVERSITY DATASET

ID	Land Cover Class	Training	Test
C01	Asphalt	34	6567
C02	Meadows	94	18525
C03	Gravel	11	2058
C04	Trees	16	3018
C05	Painted metal sheets	7	1308
C06	Bare Soil	25	4974
C07	Bitumen	7	1293
C08	Self-Blocking Bricks	19	3633
C09	Shadows	5	912
	Total	213	42293



(a) (b)

Fig. 7. Salinas dataset. (a) False-color map. (b) Ground-truth map.

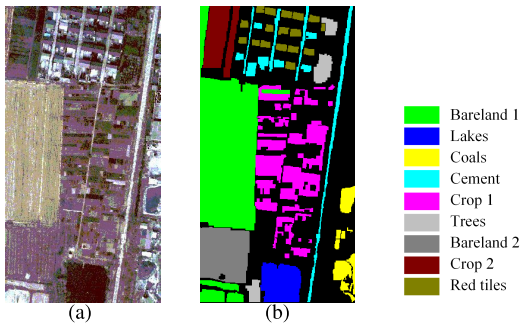
2) Salinas: The Salinas dataset was imaged by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor as a set of HSI data in the Salinas Canyon region. It has an image size of 512 × 217 pixels and contains a total of 204 bands available for classification. The specific label categories and sample set distribution have been shown in Fig. 7 and Table II.

3) Xuzhou: The Xuzhou dataset was taken in an urban and rural area of Xuzhou City China. The spatial size of the image is 500 × 260 pixels, with a very high spatial resolution of 0.73 m. In addition, the Xuzhou dataset contains 436 spectral bands. The specific label categories and sample set distribution have been shown in Fig. 8 and Table III.

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TABLE II
TRAINING AND TEST SAMPLES IN THE SALINAS DATASET

ID	Land Cover Class	Training	Test
C01	Alfalfa	11	1998
C02	Corn-notill	19	3707
C03	Corn-mintill	10	1966
C04	Corn	7	1387
C05	Grass-pasture	14	2664
C06	Grass-trees	20	3939
C07	Grass-pasture-mowed	18	3561
C08	Hay-windrowed	57	11214
C09	Oats	32	6171
C10	Soybean-notill	17	3261
C11	Soybean-mintill	6	1062
C12	Soybean-clean	10	1917
C13	Wheat	5	911
C14	Woods	6	1064
C15	Vinyard_untrained	37	7231
C16	Vinyard_vertical_trellis	10	1797
	Total	279	53850



(a) (b)

Fig. 8. Xuzhou dataset. (a) False-color map. (b) Ground-truth map.

TABLE III
TRAINING AND TEST SAMPLES IN THE XUZHOU DATASET

ID	Land Cover Class	Training	Test
C01	Bareland 1	132	26264
C02	Lakes	21	4006
C03	Coals	14	2769
C04	Cement	27	5187
C05	Crops 1	66	13118
C06	Trees	13	2423
C07	Bareland 2	35	6955
C08	Crop 2	24	4753
C09	Red Tiles	16	3054
	Total	348	68529

B. Experimental Setting

1) Evaluation Criteria: To accurately analyze and evaluate the difference in classification performance between our proposed method and other comparative methods, we evaluate the classification results using four types of metrics: overall accuracy (OA), average accuracy (AA), kappa coefficient, and classification accuracy for each category. The higher value of these four evaluation metrics indicates the better classification performance of the method.

2) Environment Configuration: Our proposed method is carried out in the PyTorch environment. For the classical traditional methods used for comparison experiments, we have conducted them all in the MATLAB R2018b environment.

TABLE IV
ABLATION EXPERIMENTAL RESULTS

Cases	Component				Indicators		
	CFAF	CSE-A	EFA-Tokenizer	Lw-Trans	OA(%)	AA(%)	$k*100$
1	×	✓	✓	✓	96.61	97.61	96.56
2	✓	×	✓	✓	97.09	97.88	96.78
3	✓	✓	×	✓	86.57	84.44	85.01
4	✓	✓	✓	×	96.82	97.61	96.46
5	✓	✓	✓	✓	**97.49**	**98.52**	**97.21**

We used an Intel Xeon Silver 4210 CPU with 128 GB of RAM and an NVIDIA Geforce RTX 2080ti 11 GB GPU server. For the deep learning and transformer-based comparison methods, the raw learning rate was set to $1e^{-3}$ and the size of each batch was set to 64.

3) Parameter Setting Adjustment: In the parameter analysis section, we analyze and compare several classes of parameters that have an impact on the classification results and adjust them to the optimal ones, including the size of the patch $s \times s$ for image partitioning, the number of tokens t, and the size of the small scale pooling layer p in pooling mixer. We conducted parameter experiments on the three datasets with an ultrasmall sample rate of 0.5% as described above.

Fig. 9 shows the effect of the number of tokens and the size of the pooling mixer on the classification accuracy. Fig. 9(a) shows the results on the PaviaU dataset. It is observed that the variation of token number versus OA coordinates, and the OA increased up to the highest when the number of tokens was 4. Also, the best classification result was obtained when the pooling size was 5×5. Fig. 9(b) shows the results on the Salinas dataset; it is found that the best result was achieved when the number of tokens was 6. Then, we also observed the variation of pooling size versus OA coordinates, and the best classification result was achieved when the pooling size was 5×5. Finally, Fig. 9(c) shows the results on the Xuzhou dataset. Based on the relationship between the number of tokens, pool size, and OAs, it shows that the best classification result was obtained when the number of tokens was 6 and the pooling size was 5×5.

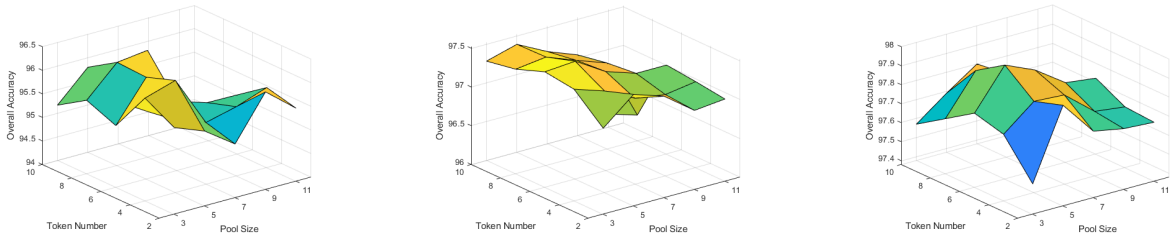
Fig. 10 shows the relationship between patch size and classification accuracy on the three datasets. A comprehensive comparison of the three subplots shows that optimal classification results were achieved on all three datasets when the patch size was 13.

C. Ablation Study

To adequately demonstrate the specific impact of each component of the proposed MAR-LWFormer on the classification performance, we conducted an ablation experiment with a sample rate of 0.5% on the Salinas dataset. We split the proposed method into four sections: CFAF, CSE-A, EFA-Tokenizer, and Lw-Trans. OA, AA, and Kappa metrics are employed to evaluate the impact of each of the four sections on the final classification results. The results of the ablation experiments are shown in Table IV.

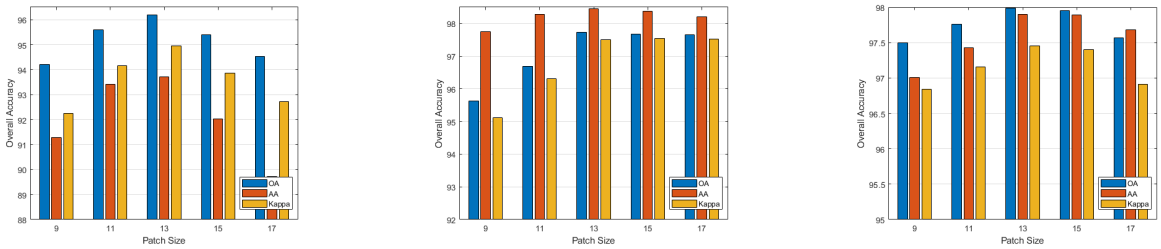
Case 1 eliminates the CFAF module and directly inputs the original unique-convolution features into the network for classification without using EMAP and LBP information for spatial feature enhancement. At this time, the spatial feature

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(a) (b) (c)

Fig. 9. Influence of impact between different numbers of tokens and pool size for the classification accuracy. (a) Pavia University dataset. (b) Salinas dataset. (c) Xuzhou dataset.



(a) (b) (c)

Fig. 10. Influence of impact on patch size for the OA, AA, and Kappa. (a) Pavia University dataset. (b) Salinas dataset. (c) Xuzhou dataset.

representation capability is weakened, and the evaluation metrics of all three types of classification accuracies are lower than the proposed MAR-LWFormer method. This illustrates the effectiveness of the CFAF module for improving the final classification accuracy.

Case 2 eliminates the CSE-A module and does not perform cross-dimensional attention enhancement on the spectral information, so the coupling of spectral–spatial features is lower than our proposed method. It is clear that all three classification accuracy metrics are lower than the proposed MAR-LWFormer. This indicates that CSE-A can effectively consider the spectral-spatial features simultaneously and enhance the effective coupling of spectral-spatial features.

Case 3 eliminates the EFA-Tokenizer module and inputs the spectral information-enhanced features directly into the transformer encoder without flattening it with the explicit feature attention, and the input data are redundant and complex. Therefore, in this case, the three classification accuracy metrics are lower than the proposed MAR-LWFormer by a relatively large value.

Case 4 does not employ the transformer encoder and uses the softmax classifier to classify the feature data directly. Definitely, all three classification metrics are lower than the proposed MAR-LWFormer, with the value of AA decreasing by the largest value. This indicates that it is difficult for the common classifier to achieve globally uniform modeling without using the transformer, especially in categories with small sample sizes. It also shows that the lightweight pooling transformer we employed also has a greater advantage in the distribution balance of samples.

In addition, we conducted further comparative ablation studies for the transformer encoder with a sample rate of 0.5% on the Salinas dataset, and the experimental results are shown in Table V. Both proposed lightweight pooling transformer and multihead self-attention transformer have extremely close values in all three types of accuracy metrics, which are higher than the ordinary softmax classifier. Also, the proposed lightweight pooling transformer has the fastest running speed during the test. The number of floating-point operations/s (FLOPs) is a metric used to measure the computational complexity of a machine learning model, indicating the number of floating-point operations required to execute the model during inference. Our proposed method shows an improvement in FLOPs compared to both the MSA-transformer and softmax methods. Additionally, our method reduces the number of model parameters by 3.73% compared to the MSA-transformer and 10.51% compared to the softmax method. Table V verifies that our proposed MAR-LWFormer has a stronger accuracy-time ratio and excellent classification capability.

TABLE V

TRANSFORMER ABLATION EXPERIMENTAL RESULTS (OPTIMAL RESULTS ARE BOLDED, AND SUBOPTIMAL RESULTS ARE UNDERLINED)

Methods	Accuracy Indicators		
	OA(%)	AA(%)	$k * 100$
Softmax	96.82±0.59	97.61±0.56	96.46±0.66
MSA-Transformer	97.44±0.31	98.31±0.31	97.15±0.34
Light-Transformer	**97.49±0.40**	**98.52±0.19**	**97.21±0.44**
Methods	Efficiency Indicators		
	Test Time(s)	FLOPs(GB)	Parameters(KB)
Softmax	16.62±0.25	1.85	359.86
MSA-Transformer	17.55±1.04	1.84	334.51
Light-Transformer	<u>16.15±0.28</u>	<u>1.83</u>	<u>322.03</u>

D. Classification Results

To quantitatively verify that our proposed method performs efficient HSI classification at ultrasmall sample rates, we still compare MAR-LWFormer with those state-of-the-art methods including RF, SVM, 2D-CNN, 3D-CNN, HybridSN, MSAF-KELM, SPRLT-Net, and SSFTT.

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TABLE VI

CLASSIFICATION PERFORMANCE OF PAVIA UNIVERSITY DATASET OBTAINED BY DIFFERENT METHODS (THE BEST RESULTS ARE SHOWN IN BOLD)

No.	Traditional Methods		Deep Learning Networks						
	Random Forest [41]	SVM [42]	2D-CNN [43]	3D-CNN [44]	HybridSN [27]	MSAF-KELM [31]	SPRLT-Net [38]	SSFTT [39]	MAR-LWFormer
1	81.66±7.31	82.27±3.95	90.52±2.97	92.41±5.92	88.76±3.21	93.71±0.37	96.45±1.63	90.70±7.98	**97.53±1.69**
2	93.59±2.76	93.56±1.89	98.68±0.99	99.16±1.29	98.98±0.62	99.68±0.25	99.76±0.29	98.87±1.35	**99.94±0.06**
3	33.36±12.33	60.50±9.59	73.00±6.79	60.45±11.02	68.89±3.03	**85.00±5.31**	71.45±6.20	78.55±8.29	83.56±1.20
4	69.95±7.53	81.66±2.75	79.51±6.94	78.76±4.79	74.51±8.74	86.73±4.55	90.01±2.63	83.91±3.69	**90.1±3.75**
5	97.32±2.45	94.28±11.37	99.67±0.75	99.95±0.10	92.76±6.51	96.56±0.80	99.98±0.04	**99.99±0.03**	99.98±0.07
6	36.04±7.18	71.53±3.68	90.67±1.72	87.21±6.89	73.80±8.56	**98.55±1.03**	91.79±6.63	96.42±1.99	95.12±2.85
7	55.22±14.34	72.60±13.77	90.21±8.44	97.86±2.64	94.65±2.43	81.76±7.31	90.23±8.70	**99.87±0.15**	96.15±4.33
8	76.29±7.55	79.08±4.16	73.54±4.89	75.30±7.71	78.77±5.81	**93.83±2.63**	81.41±2.64	79.88±6.31	89.02±2.82
9	99.25±0.29	**99.61±0.20**	85.55±10.29	85.55±7.42	62.03±19.13	54.88±9.06	58.49±25.72	90.35±5.65	95.42±2.61
OA(%)	77.89±1.03	85.00±1.39	91.15±0.47	90.98±1.13	88.32±2.21	94.82±0.31	93.44±0.58	93.49±1.64	**96.34±0.35**
AA(%)	71.41±2.10	81.68±2.35	86.82±1.10	86.29±1.80	81.46±4.59	87.86±1.85	86.62±3.32	90.95±1.34	**94.09 ±0.57**
k×100	69.83±1.33	79.98±1.85	88.17±0.63	87.91±1.53	84.24±3.08	93.10±0.41	89.86±1.65	91.22±0.81	**95.12±0.47**

TABLE VII

CLASSIFICATION PERFORMANCE OF SALINAS DATASET OBTAINED BY DIFFERENT METHODS (THE BEST RESULTS ARE SHOWN IN BOLD)

No.	Traditional Methods		Deep Learning Networks						
	Random Forest [41]	SVM [42]	2D-CNN [43]	3D-CNN [44]	HybridSN [27]	MSAF-KELM [31]	SPRLT-Net [38]	SSFTT [39]	MAR-LWFormer
1	97.70±1.49	95.98±2.60	98.08±4.98	**99.99±0.02**	98.55±1.57	98.50±0.74	98.57±2.75	99.87±0.19	99.8±0.40
2	99.34±1.06	95.01±1.20	**100.00±0.00**	**100.00±0.00**	93.27±9.30	99.10±0.57	**100.00±0.00**	99.98±0.03	**100.00±0.00**
3	89.47±6.06	46.50±11.91	99.52±0.35	99.41±1.61	96.34±1.15	99.59±0.38	99.16±1.42	**100.00±0.00**	99.99±0.02
4	93.47±4.63	96.84±1.00	91.94±9.39	94.54±2.17	89.83±2.54	92.29±2.21	92.10±7.94	95.60±3.52	**98.23±1.23**
5	94.57±2.41	97.37±1.05	99.84±0.46	99.75±0.37	93.61±7.29	95.82±2.45	97.77±1.98	98.62±0.26	**99.52±0.68**
6	97.65±1.90	98.53±0.60	99.19±1.14	99.94±0.09	99.37±0.83	98.87±0.63	**100.00±0.00**	99.82±0.07	99.93±0.09
7	98.27±0.81	99.30±0.10	99.88±0.26	99.88±0.14	98.39±1.49	99.21±0.85	**99.96±0.07**	99.02±0.62	99.51±0.52
8	79.52±3.51	93.98±5.36	87.03±4.71	88.70±3.56	94.20±0.76	97.41±0.20	91.99±3.82	92.36±2.03	95.18±1.56
9	99.05±0.28	99.02±0.12	99.99±0.02	99.96±0.09	99.86±0.12	99.98±0.02	**100.00±0.00**	99.97±0.03	99.98±0.03
10	72.81±5.23	72.38±5.87	97.35±2.02	98.64±0.83	95.60±3.96	93.89±2.33	96.77±1.66	97.95±1.23	**98.99±0.38**
11	74.96±7.41	0.27±0.77	98.34±2.33	99.56±0.41	94.92±3.46	88.02±5.41	99.82±0.16	**99.97±0.04**	99.62±0.48
12	93.94±5.92	96.85±1.40	98.47±2.29	99.62±0.35	86.70±4.89	97.39±2.59	**99.68±0.52**	96.42±3.48	99.54±0.54
13	96.56±5.32	**98.65±0.36**	93.07±9.79	87.00±12.86	91.07±1.45	88.91±6.35	72.99±29.70	92.65±6.69	98.2±1.19
14	89.83±3.02	87.65±2.60	85.08±26.64	97.35±1.17	83.69±18.62	93.05±3.86	93.13±8.34	95.99±2.53	**97.41±1.60**
15	50.03±7.50	9.75±12.97	86.66±3.54	87.60±2.45	89.24±5.93	**92.86±1.18**	85.48±6.82	89.76±1.51	91.04±1.73
16	83.40±8.40	66.15±4.48	99.13±0.40	99.12±0.10	96.98±2.62	93.96±4.97	98.89±0.78	99.33±0.25	**99.43±0.29**
OA(%)	84.51±0.92	78.55±0.83	94.45±0.79	95.38±0.70	94.54±3.01	96.60±0.25	95.14±1.03	96.29±0.66	**97.49±0.40**
AA(%)	88.16±1.22	78.39±1.06	95.85±1.87	96.94±0.93	93.85±3.78	95.55±0.44	95.39±1.90	97.33±0.23	**98.52±0.19**
k×100	82.70±1.05	75.80±1.01	93.82±0.88	94.86±0.77	93.92±3.36	96.21±0.28	94.59±1.15	95.87±0.74	**97.21±0.44**

TABLE VIII

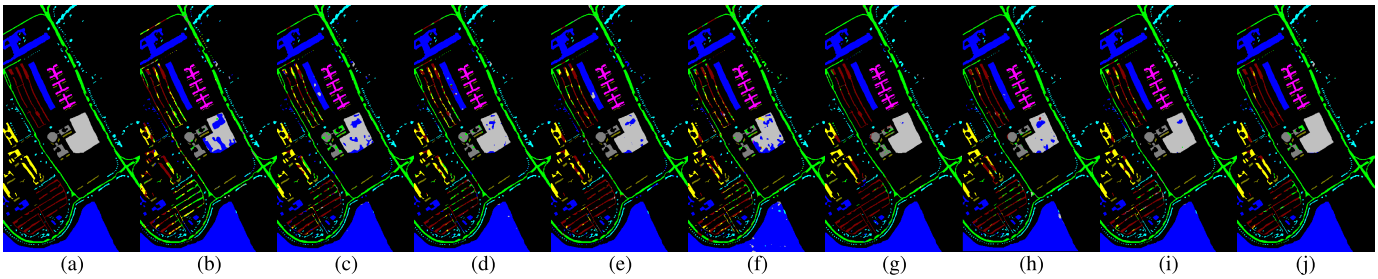
CLASSIFICATION PERFORMANCE OF XUZHOU DATASET OBTAINED BY DIFFERENT METHODS (THE BEST RESULTS ARE SHOWN IN BOLD)

No.	Traditional Methods		Deep Learning Networks						
	Random Forest [41]	SVM [42]	2D-CNN [43]	3D-CNN [44]	HybridSN [27]	MSAF-KELM [31]	SPRLT-Net [38]	SSFTT [39]	MAR-LWFormer
1	92.51±1.19	93.78±1.29	96.48±0.59	95.69±2.34	94.40±0.82	97.00±0.45	**97.67±0.48**	95.78±2.19	97.61+0.39
2	95.54±1.90	96.73±0.33	96.47±0.73	96.85±0.90	96.17±1.86	**98.99±0.80**	98.78±0.54	97.58±1.21	97.76+0.84
3	69.50±10.09	73.17±7.22	81.40±4.71	88.28±3.81	91.06±5.53	94.55±1.50	91.38±4.98	93.45±3.10	**97.39+3.28**
4	78.87±6.19	85.21±3.35	96.81±1.54	97.23±1.14	86.43±8.19	83.39±1.74	96.59±2.03	**99.18±0.48**	98.56+2.01
5	88.93±2.10	94.05±2.11	96.12±1.07	97.62±1.28	95.82±2.07	94.75±2.30	**98.80±0.34**	97.35±2.06	98.50+0.45
6	54.31±5.03	39.38±10.75	95.91±3.15	91.26±3.90	81.17±13.09	93.69±2.23	92.79±5.07	96.84±2.24	**99.08+1.38**
7	73.28±4.23	61.71±5.78	98.21±0.88	97.40±1.02	97.44±0.87	96.70±1.99	98.74±0.65	**99.62±0.41**	98.69+0.54
8	95.91±0.50	95.73±0.69	99.39±0.82	98.29±0.81	99.40±0.40	95.97±1.23	97.18±2.17	99.69±0.30	**99.93+0.14**
9	79.86±9.66	72.05±6.78	95.83±2.69	93.59±2.70	96.52±1.40	84.70±5.05	**98.54±1.34**	92.58±3.32	93.44+1.62
OA(%)	86.41±1.06	86.51±0.58	96.16±0.30	96.05±0.96	94.32±1.26	94.79±0.45	97.56±0.30	96.90±1.35	**97.99±0.19**
AA(%)	80.97±1.92	79.09±1.56	95.18±0.72	95.13±0.85	92.16±1.60	93.30±0.50	96.72±0.80	96.90±0.92	**97.88±0.36**
k×100	82.64±1.37	82.63±0.76	95.13±0.38	95.01±1.19	92.82±1.62	93.37±0.57	96.90±0.38	96.10±1.68	**97.45±0.24**

Tables VI–VIII show the experimental results of MAR-LWFormer and several comparison methods on three different datasets with a sampling rate of 0.5%. To ensure the validity and generality of the experimental results, we repeated each experiment ten times and then took their mean and standard deviation as the final results.

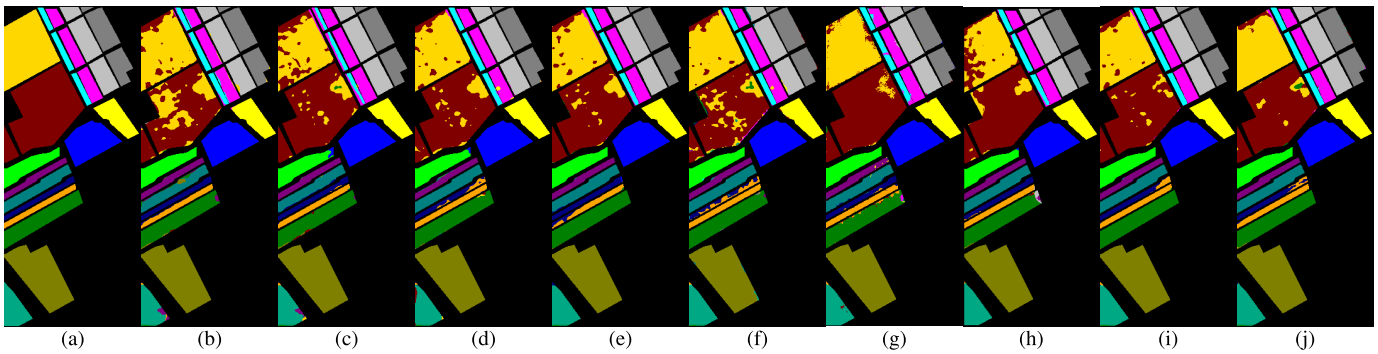
First, for the Pavia University dataset, the results are shown in Table VI. Our method achieved the highest performance for all classification accuracies, as well as for OA, AA, and Kappa values. Compared to traditional methods such as RF and SVM, the deep learning methods generally had higher classification accuracies. The 2D-CNN, 3D-CNN, and HybridSN methods extracted the most standard convolution features, but

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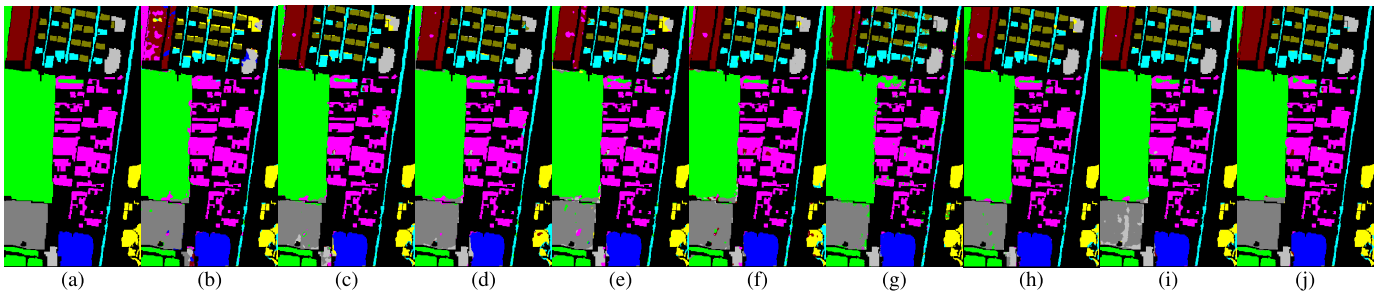
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 11. Classification maps for Pavia University dataset under different methods. (a) GroundTruth. (b) RF. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HybridSN. (g) MSAF-KELM. (h) SPRLT-Net. (i) SSFTT. (j) MAR-LWFormer.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 12. Classification maps for Salinas dataset under different methods. (a) GroundTruth. (b) RF. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HybridSN. (g) MSAF-KELM. (h) SPRLT-Net. (i) SSFTT. (j) MAR-LWFormer.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 13. Classification maps for Xuzhou dataset under different methods. (a) GroundTruth. (b) RF. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HybridSN. (g) MSAF-KELM. (h) SPRLT-Net. (i) SSFTT. (j) MAR-LWFormer.

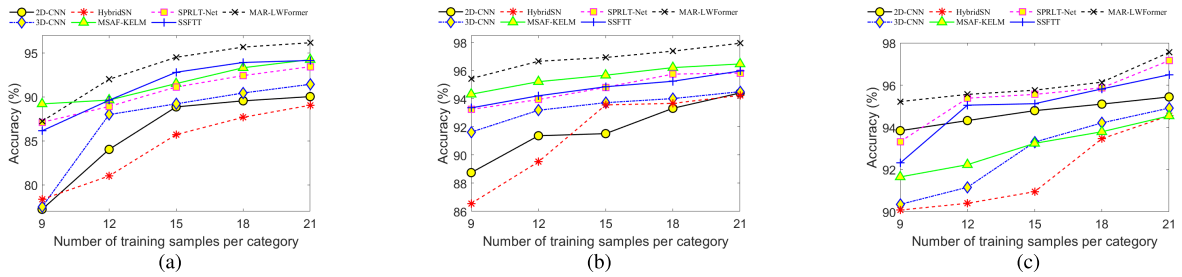


Fig. 14. Influence of different training sample sizes of 2D-CNN, 3D-CNN, HybridSN, MSAF-KELM, SPRLT-Net, SSFTT, and MAR-LWFormer methods on classification accuracy. (a) Pavia University dataset. (b) Salinas dataset. (c) Xuzhou dataset.

their OA results were 5.19%, 5.36%, and 8.02% lower than MAR-LWFormer, respectively, due to their focus on local information at the expense of global wholeness. The MSAF-KELM method, which used an attention strategy for KELM deep network fusion, had OA, AA, and Kappa accuracies that were 1.52%, 6.23%, and 2.02% lower than those of MAR-LWFormer, respectively. It is worth emphasizing that MAR-LWFormer achieved higher AA than MSAF-KELM by a relatively large value, which indicated that our method has a better sample balance ability due to the improvement on the coupling of spatial–spectral features in the case of small training samples. For instance, even for some categories with particularly small sample sizes, such as *Bitumen* and *Shadows*, our method still achieved relatively high accuracies. Transformer-based deep networks, such as SPRLT-Net and SSFTT, had better OA performance because they both used

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a multihead self-attention mechanism for modeling and made full use of global feature information. However, since the MAR-LWFormer method enhanced both local spectral-spatial features and utilized a pooling mixer to better capture the global information, it had higher OA, AA, and Kappa accuracies than those of SPRLT-Net and SSFTT.

Table VII shows the quantitative experimental results on the Salinas dataset at 0.5% sample rate. On this dataset, we still had the highest accuracies among all compared methods. Salinas dataset has a low spatial resolution, therefore, the CFAF module of MAR-LWFormer at an ultrasmall sample rate effectively achieves the enhancement of spatial information, reflecting the advantage of multifeature information interaction. Compared with 2D-CNN, 3D-CNN, and HybridSN methods, our method achieved better performance in terms of OA and AA because our method utilized both local and non-local features. For the MSAF-KELM method, MAR-LWFormer still outperformed MSAF-KELM in sample balance processing with 2.97% higher average precision because our method made full use of EMAP features and LBP features for spatial feature enhancement. For SPRLT-Net and SSFTT methods, since the proposed MAR-LWFormer used the EFA-Tokenier module for redundant features rejection of token features, which effectively improved the classification efficiency of the transformer encoder, it was more competitive than SPRLT-Net and SSFTT in OA, AA, and Kappa accuracies.

Finally, the results of the Xuzhou dataset are shown in Table VIII. The Xuzhou dataset is the highest spatial resolution dataset in our experiment, reaching 0.73 m. Also, it has the largest number of spectral bands, reaching 436. For the comprehensive evaluation metrics of OA and Kappa, the performance of our proposed method was the best. Due to the rich information, the transformer encoder had sufficient global information for modeling, and both transformer-based networks achieved high classification accuracy. However, due to the pooling mixer used by MAR-LWFormer, it can better achieve global information interaction and enhance the feature representation of imbalanced samples. Therefore, for both *Coals* category and *Red Tiles* category, our method had higher classification accuracies. Numerically, for the AA, our method was 1.16% and 0.98% higher than SPRLT-Net and SSFTT, respectively.

E. Visual Evaluation

The classification maps of the proposed method MAR-LWFormer on three datasets with 0.5% sample rate have been shown in Figs. 11–13. By comparing the region shape, edge contour, and noise range of the maps, the visual maps of the MAR-LWFormer method were the closest to the Ground-Truth maps. The results of traditional classifiers were generally worse than those of deep learning classifiers, and their classification maps presented a higher amount of misclassification phenomena accompanied by severe noise. For 2D-CNN, 3D-CNN, and HybridSN methods, the misclassification phenomenon was reduced, but there were still obvious noise outliers. For the MSAF-KELM method, the noise problem was greatly alleviated in the classification map, and the shape of the classified region was accurate. For the transformer-based methods, e.g., SPRLT-Net and SSFTT, the classification accuracy was further improved and the outline of the classification map was clear because the global information interaction was taken into account. Finally, for the classification map of the proposed method MAR-LWFormer, it was the one with the least misclassification cases and noise phenomenon, which also showed that our method had the best visual classification performance at a small sample rate.

Specifically, in the Pavia University dataset, RF, SVM, 2D-CNN, 3D-CNN, HybridSN, and SPRLT-Net all appeared to misclassify some of the *Bare_Soil* (in gray) categories as *Meadows* (in blue) categories. In addition, MSAF-KELM and SSFTT methods also showed noise in the *Bare_Soil* (in Gray) category. Only MAR-LWFormer had the most accurate classification results here. Second, for the Salinas dataset, two traditional methods presented a significant amount of misclassification phenomena in the *Vinyard_untrained* (in dark-yellow) and *Grapes_untrained* (in red-brown) category regions. 2D-CNN, 3D-CNN, and HybridSN networks and MSAF-KELM method presented relatively severe pretzel noise at *Lettuce_romaine_6wk* (in blue) category and *Lettuce_romaine_wk* (in orange) category. SPRLT-Net and SSFTT only appeared a small quantity of misclassification in *Grapes_untrained* (in red-brown) category region. MAR-LWFormer had the least classification error in the *Grapes_untrained* (in red-brown) category region among all methods. Third, for the Xuzhou dataset, all methods showed dotted classification noise in the *Bareland_2* (in dark gray) category, and only MAR-LWFormer's classification map maintained the most accurate and clear classification result in this category.

In conclusion, the comparison of several state-of-the-art classifiers in visual effect still validates that the proposed MAR-LWFormer has excellent classification performance.

F. Comparison of the Influence of Training Sample Size

To further evaluate the generalization ability of MAR-LWFormer, we compared it with 2D-CNN, 3D-CNN, HybridSN, MSAF-KELM, SPRLT-Net, and SSFTT using different training sample sizes. The number of samples per class was set to 9, 12, 15, 18, and 21 for the Salinas, Pavia University, and Xuzhou datasets. The experimental results are shown in Fig. 14. As can be seen from the figure, OA increased as the sample size increased, and in most cases, MAR-LWFormer maintained the highest accuracy. Additionally, MAR-LWFormer achieved the highest accuracy on both Salinas and Xuzhou datasets with sample size of 9, demonstrating its ability to extract deep spectral-spatial features even with ultrasmall samples. The slightly lower accuracy on Pavia University may be due to different fusion methods under complex multifeature tasks. The rapid improvement in accuracy with a more reasonable number of training samples suggests that CFAF and CSE-A modules effectively enhance convolution feature representation through attention strategy.

IV. Conclusion

In this article, a novel MAR-LWFormer is proposed for effective HSI classification at ultrasmall sample rates. The

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method extracts original spectral, EMAP, and LBP convolved features through a three-branch network, and a CFAF module is designed to achieve enhancement of spatial information. A CSE-A module is designed to improve the coupling of spectral-spatial features. Finally, a lightweight transformer with EFA-tokenizer is used to implement classification learning. The experimental results on three datasets demonstrate the excellent classification performance of MAR-LWFormer at ultrasmall sample rates.

In the future, we will consider expanding the multifeature branches network into a multimodal branch network. It will be combined with LiDAR data to extract more abundant spectral–spatial features to improve the classification performance at ultrasmall sample rates.

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